COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	16/07/26	**DATE SUBMITTED:**	16/07/26
NAME:	Jeus Pizarro	**INSTRUCTOR:**	Engr Jimlord Quejado

ACTIVITY NO. ___ Creating Virtual Machines

1. PROCEDURE OUTPUT

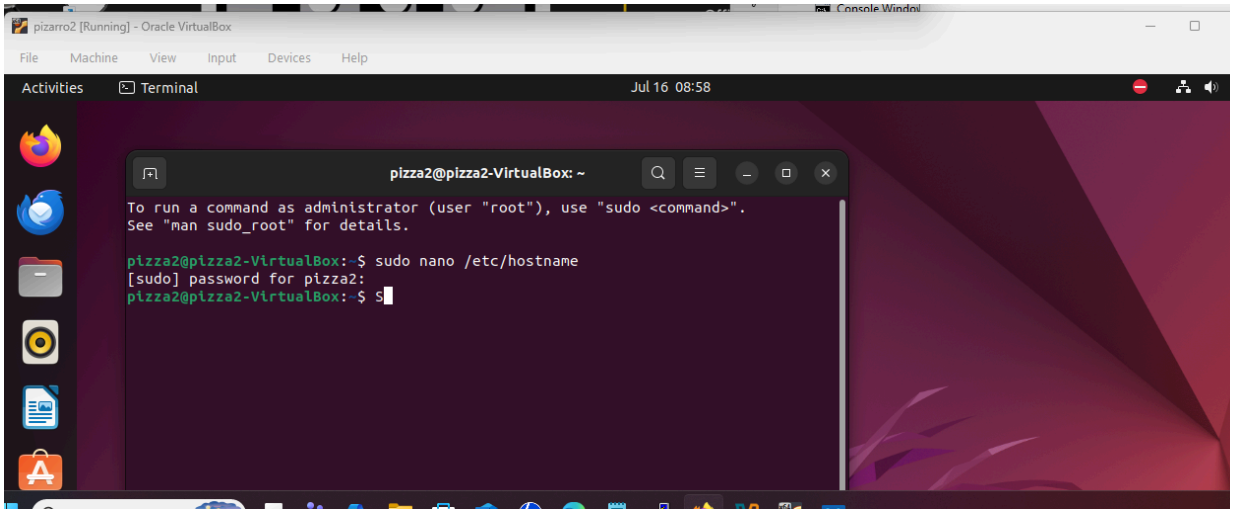
What procedures did you perform?

Task 1: Changing Hostname

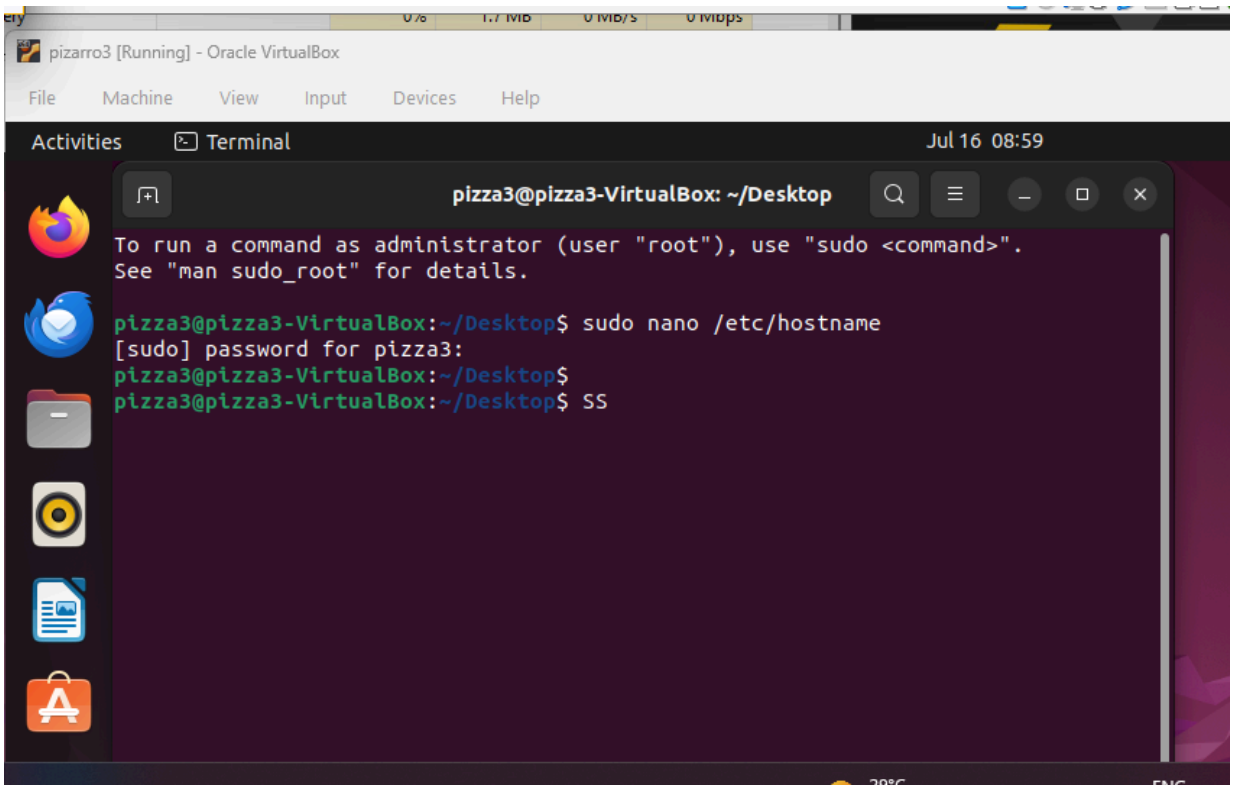
Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press Control + O to write out (save the file). Press Enter when asked for the name of the file. Press Control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

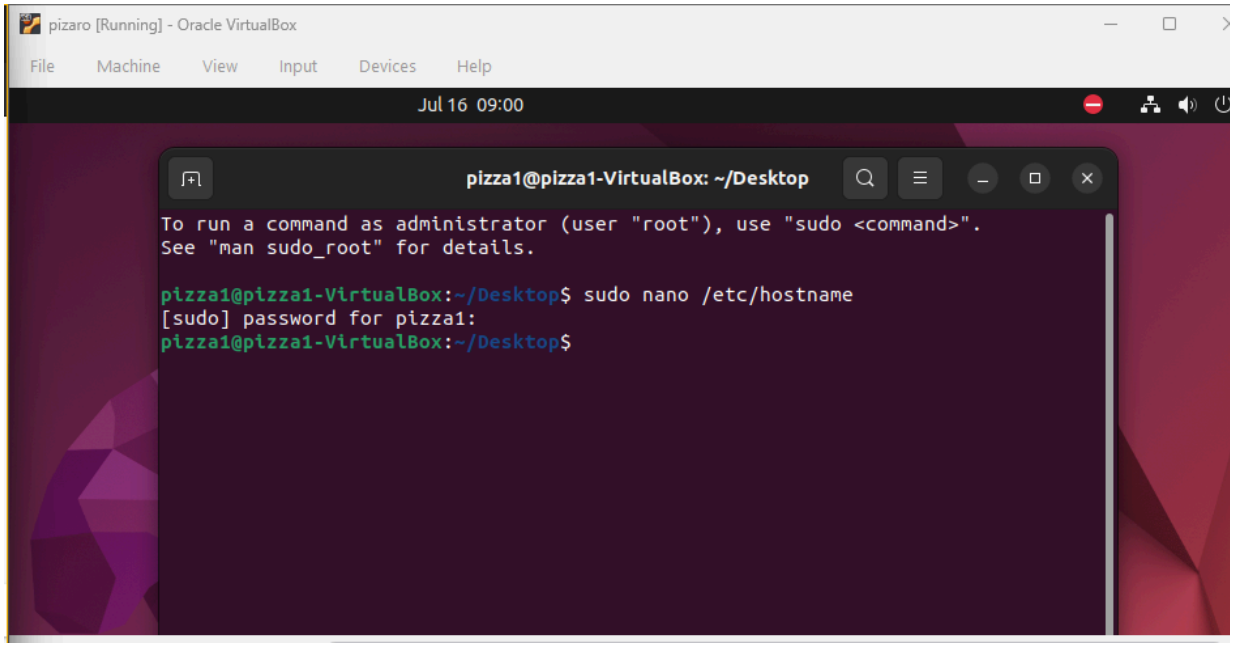
1.1 Use server1 for Server 1



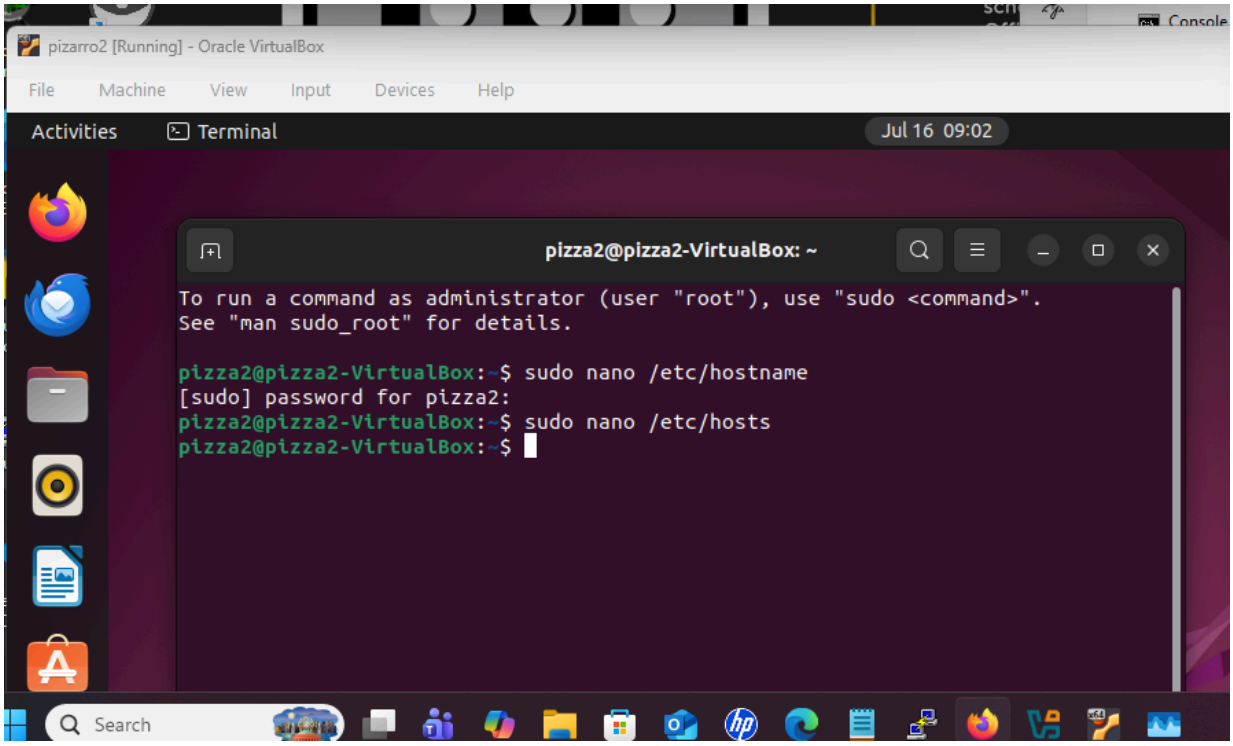
1.2 Use server2 for Server 2



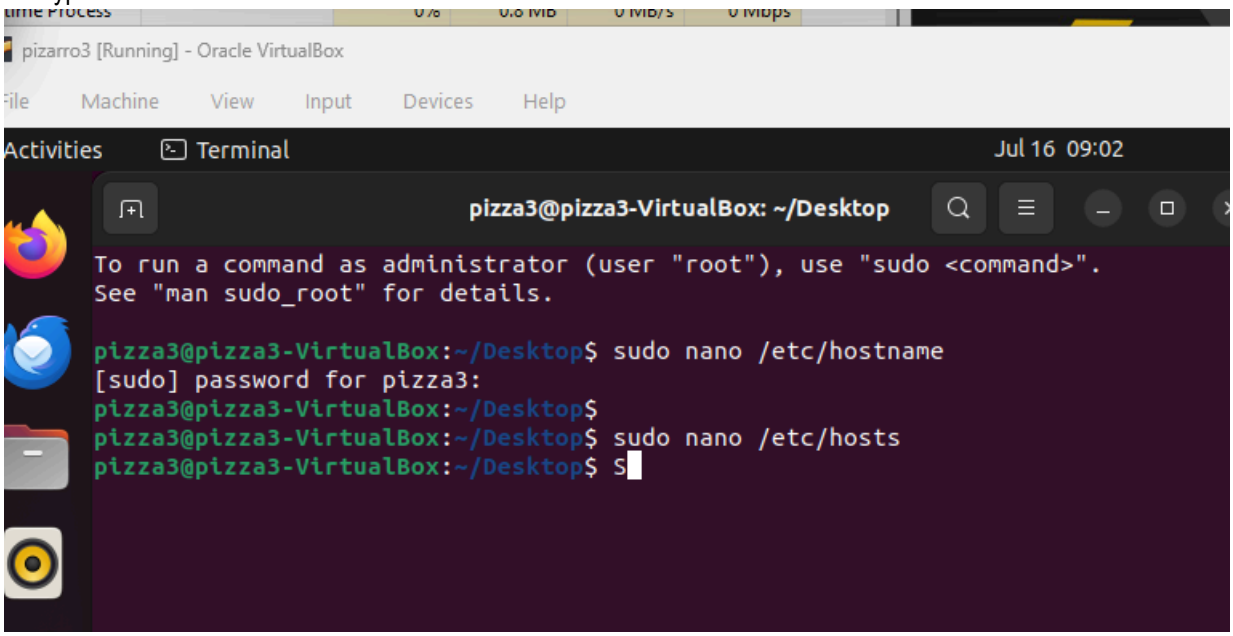
1.3 Use workstation for the Local Machine



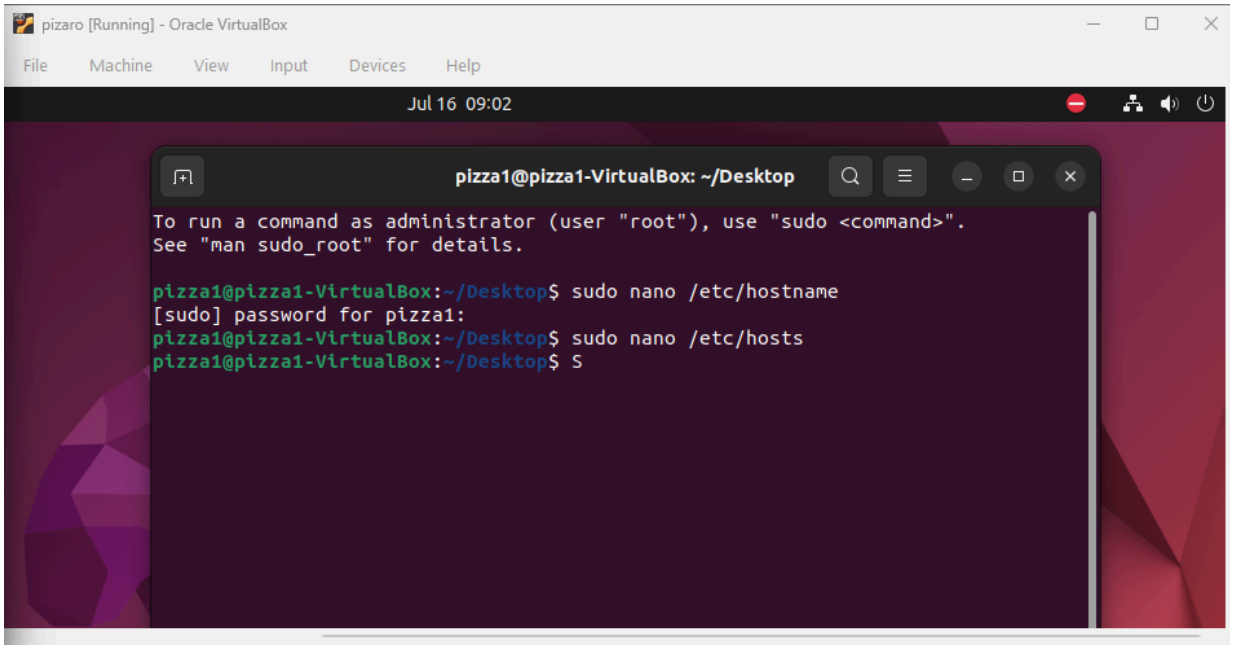
2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine



Task 2: Configure SSH

Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the commands sudo apt update and sudo apt upgrade, respectively.

```
pizza2@pizza2-VirtualBox:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Fetched 129 kB in 1s (86.8 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

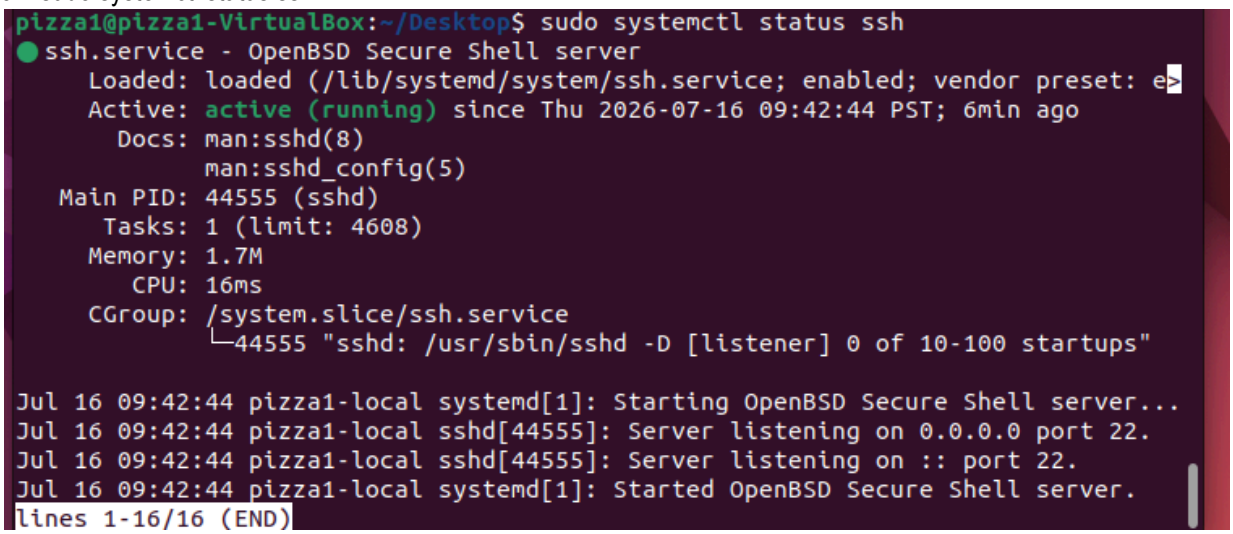
2. Install the SSH server using the command sudo apt install openssh-server.

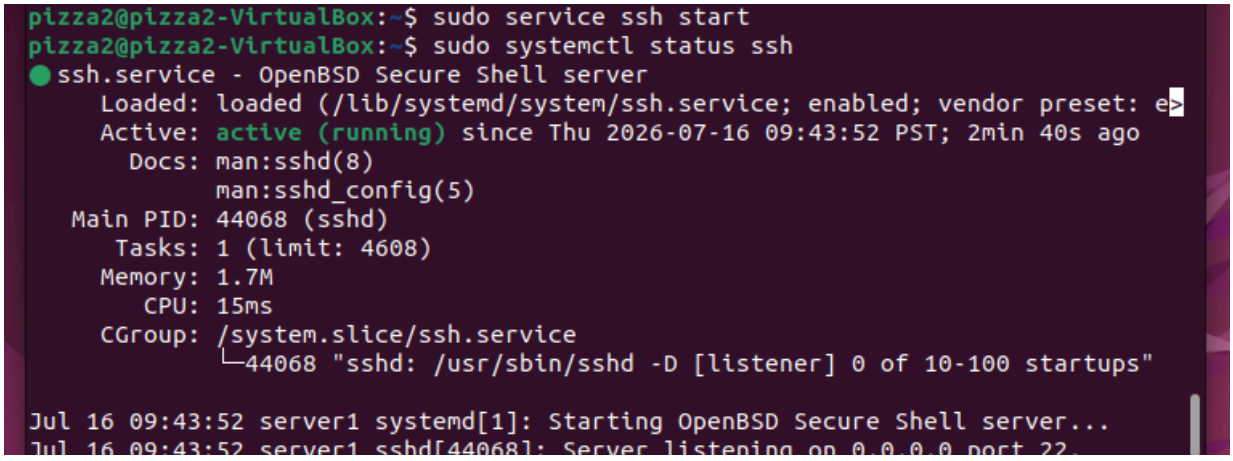
```
pizza2@pizza2-VirtualBox:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
```

3. Verify if the SSH service has started by issuing the following commands:

3.1 sudo service ssh start

3.2 sudo systemctl status ssh

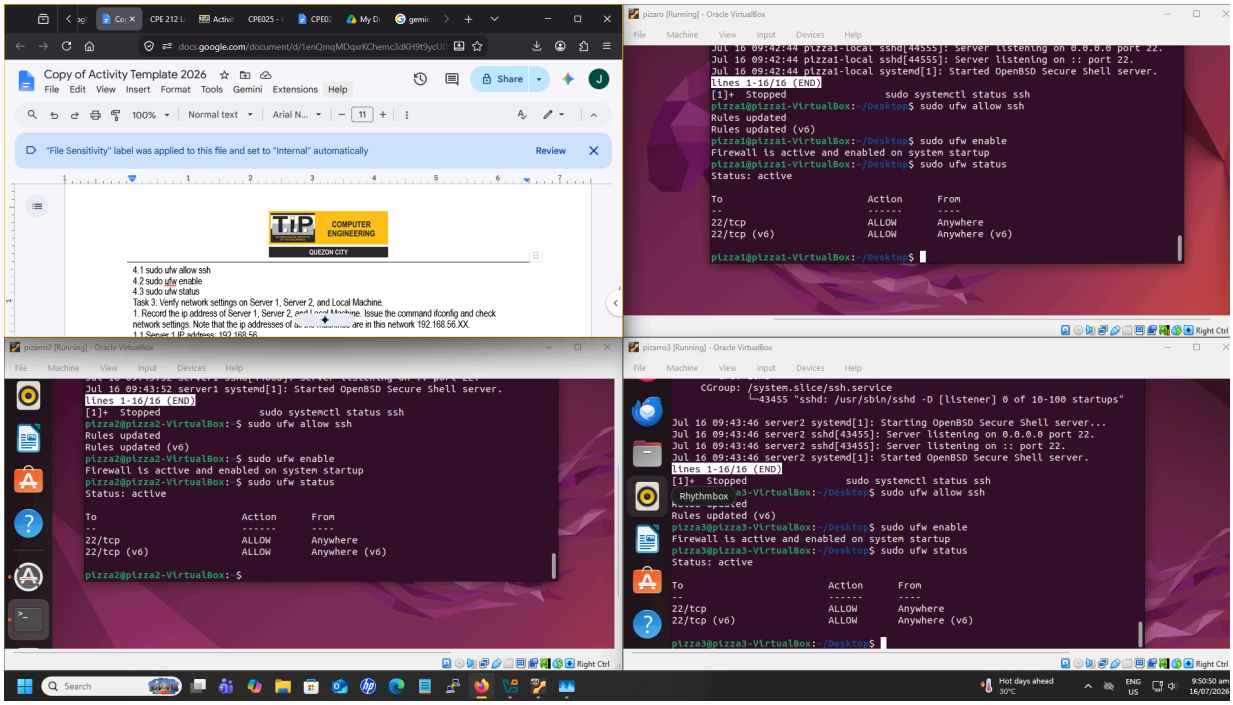






4. Configure the firewall to all port 22 by issuing the following commands:

4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status



Task 3: Verify network settings on Server 1, Server 2, and Local Machine.

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.110

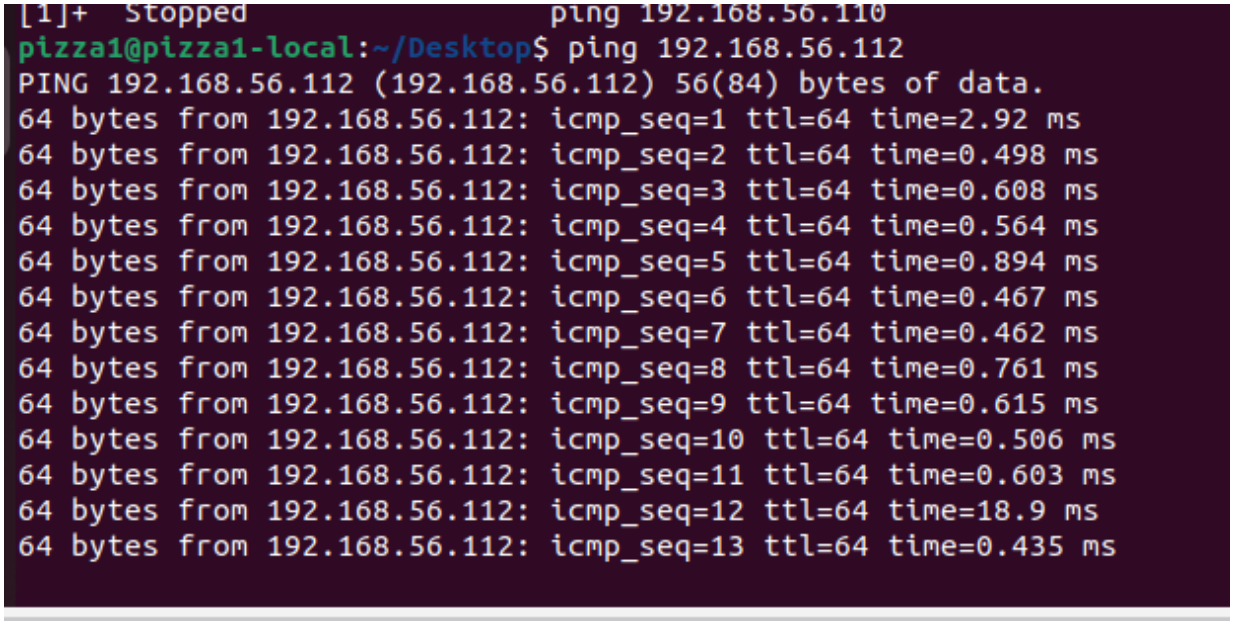
1.2 Server 2 IP address: 192.168.56.112

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☐ Successful

```
pizza1@pizza1-local:~/Desktop$ ping 192.168.56.110
PING 192.168.56.110 (192.168.56.110) 56(84) bytes of data.
64 bytes from 192.168.56.110: icmp_seq=1 ttl=64 time=1.80 ms
64 bytes from 192.168.56.110: icmp_seq=2 ttl=64 time=0.461 ms
64 bytes from 192.168.56.110: icmp_seq=3 ttl=64 time=0.439 ms
64 bytes from 192.168.56.110: icmp_seq=4 ttl=64 time=0.544 ms
64 bytes from 192.168.56.110: icmp_seq=5 ttl=64 time=0.486 ms
64 bytes from 192.168.56.110: icmp_seq=6 ttl=64 time=0.413 ms
64 bytes from 192.168.56.110: icmp_seq=7 ttl=64 time=0.456 ms
64 bytes from 192.168.56.110: icmp_seq=8 ttl=64 time=0.453 ms
64 bytes from 192.168.56.110: icmp_seq=9 ttl=64 time=0.493 ms
64 bytes from 192.168.56.110: icmp_seq=10 ttl=64 time=0.728 ms
64 bytes from 192.168.56.110: icmp_seq=11 ttl=64 time=0.815 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: ☐ Successful



2.3 Connectivity test for Server 1 to Server 2: ☐ Successful ☐ Not Successful

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:
1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120
1.2 Enter the password for server 1 when prompted
1.3 Verify that you are in server 1. The user should be in this format user@server1.
For example, jvtaylar@server1
2. Logout of Server 1 by issuing the command control + D.
3. Do the same for Server 2.
4. Edit the hosts of the Local Machine by issuing the command sudo nano/etc/hosts. Below all texts type the following:
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
4.3 Save the file and exit.
5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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